AC 2012-3198: STUDY OF A PHOTO-VOLTAIC (PV) SYSTEM USING EXCEL: ECONOMIC ANALYSIS, MODELING, SIMULATION, AND OPTIMIZATION

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Study of a Photo-Voltaic (PV) System Using Excel: Economic Analysis, Modeling, Simulation and Optimization

Abstract

This paper is concerned with the economic analysis as well as modeling, simulation and optimization of a Photo-voltaic (PV) system by using a widely available and inexpensive computer platform. The impact of PV system location, wattage cost, loan interest rates, and government incentives on its economics is analyzed. Modeling, simulation and optimization of the PV system is accomplished using Microsoft Excel and its solver functions. A user friendly Microsoft Excel application has been developed to perform the economic analysis and system optimization. Economic analysis results clearly show that future research emphasis must be placed on developing more efficient PV system materials at lower wattage cost. The simulation and optimization results of the system are compared and are in agreement with experimental data reported for the MSX 60 PV system. Furthermore, to the best of the authors' knowledge, this is the first reported work where Microsoft Excel, an inexpensive and widely used tool, is used to model and optimize a PV system.

I. Introduction

We hear and read about many concerns for the increasing energy costs. We feel the impact of such costs increases in our day to day life. Skyrocketing oil prices, global warming, and impact to the environment are of concern to many. Fossil fuels cannot remain the dominant source of energy forever. Alternative and renewable energy sources should be sought. As a source of energy, nothing matches the sun. Only a small fraction of the sun's power output strikes the earth, but even that provides 10,000 times as much as all the commercial energy that humans use on the planet. However, only 1 percent of total energy consumption is solar, compared with roughly 85 percent from oil, natural gas, and coal[1].

Even though a number of technologies are already available, improving them to a level that leads to widespread use of solar power requires continued research and engineering efforts. This work focuses on one of these technologies, namely the photovoltaic (PV) systems.

PV systems are comprised of photovoltaic cells. A photovoltaic cell is designed to convert radiant energy in light to electrical energy that can be used to power devices. They are usually made of semiconductors, like silicon, and have an anti-reflective coating. In a photovoltaic cell, one side has a positive charge and the other side has a negative charge. When photons from light hit the cell, electrons are knocked loose. These loose electrons flow towards the positive side of the cell, creating a current. This current can now be used as a source of electricity. A simple schematic[2] is shown below in Fig. 1.

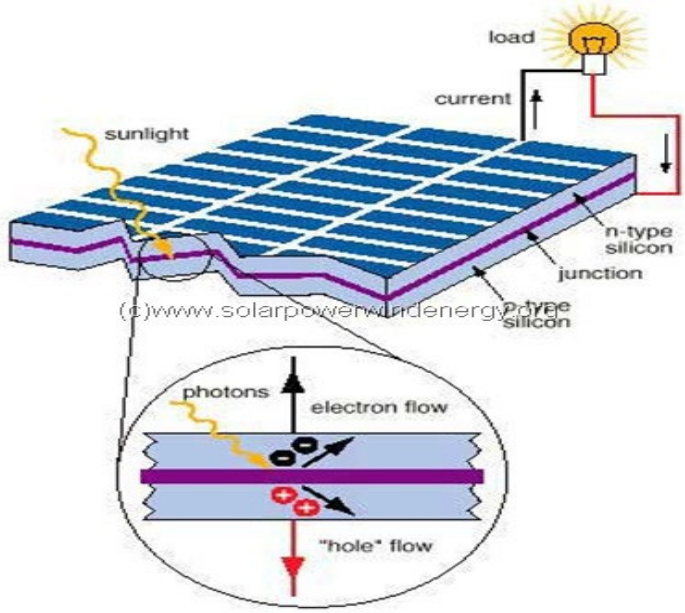


Fig. 1: PV System Schematic (source: www.solarpowerwindenergy.com)

A number of models have been developed to describe such PV cells[3-5]. They describe the current, voltage, and power output as a function of solar insolation, cell temperature, and load voltage. From a simulation viewpoint, they have been implemented using SPICE or Matlab[6]. None of them has been implemented in more widely used platforms such as Microsoft Excel.

In light of this, the objectives of this paper are to study the economics of a PV system and its operation, and develop simple, user-friendly tools to facilitate such study. The impact of PV system location, wattage cost, loan interest rates, and government incentives on its economics is analyzed. Modeling, simulation and optimization of the PV system help understand the system's operation. A user friendly Microsoft Excel application has been developed to perform the economic analysis and system optimization. The simulation and optimization results of the system are compared with experimental data reported for the MSX 60 PV system.

II. Economic Analysis – Procedure and Results

The objective is to determine the economic viability of a PV system as an alternate energy source when compared to electricity from the grid. The economic analysis will consider a number of factors such as system location, wattage cost, loan interest rates, and government incentives. The economic analysis is facilitated by developing user friendly Microsoft Excel based tools. The following procedure is used to perform the economic analysis.

Procedure

The following steps are involved:

1. Calculate the energy requirements.

   The daily amount of energy, Q, required to meet house energy needs can be read from past utility bills.

2. Estimate the Effective Sunlight Hours for the system location, H

   The following map (Fig. 2) can be used to find the effective sunlight hours for the system location[7].

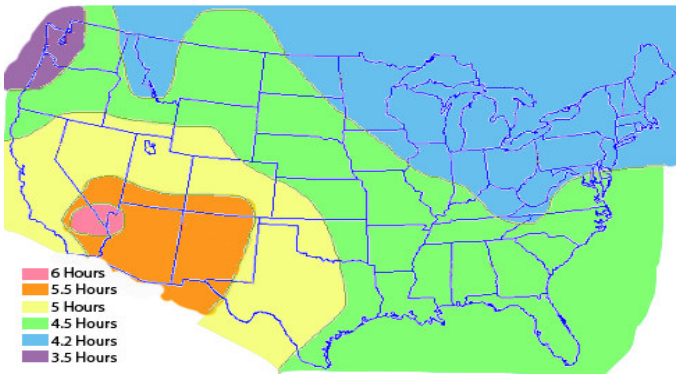


Fig. 2: Map for Effective Sunlight Hours (source: www.nrel.org)

3. Estimate the Lowest Solar Insolation for the location, I

   The following table (Table 1) has been developed for a few locations based on information reported by the National Renewable Energy Laboratory[7].

Table -1: Solar Insolation

City	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year Avg (kWh/m^2/day)
Houston	2.47	3.5	4.4	5.59	6.03	6.45	6.36	6.07	5.46	4.61	3.3	2.44	4.72
Las Vegas	3.02	4.13	5.05	6.57	7.25	7.69	7.37	6.42	6.08	4.26	3.18	2.6	5.3
Phoenix	3.25	4.41	5.17	6.76	7.42	7.7	6.99	6.11	6.02	4.44	3.52	2.75	5.38
Los Angeles	3.09	4.25	5.09	6.58	7.29	7.62	7.45	6.72	6.11	4.42	3.43	2.72	5.4
Honolulu	4.38	5.15	5.99	6.69	7.05	7.48	7.37	7.07	6.51	5.46	4.41	4.01	5.96

4. Calculate the Capacity of Solar Energy System, PV_C

   Let us assume that the energy cost for the PV system (i.e. wattage cost) is Cw. If Ep is the PV system efficiency factor and Pep is the percent of energy needs that come from the PV system, then:

$$PV_C = \frac{Q \cdot PE_P}{H \cdot E_P}$$

5. Calculate the Area of Solar Energy System, PV_A

$$PV_A = \frac{PV_C \cdot H}{I}$$

6. Calculate the Cost of Solar Energy System, C_P

   The total cost of the system includes materials and installation.

$$C_P = PV_C \cdot 1000 \cdot C_W$$

7. Calculate Actual Cost of the Solar Energy System, C_A

   The government offers a 30% tax credit off of the price of the collector. So, the actual cost of the collector can be expressed as

   $$C_A = C_P \cdot 0.7$$

8. Calculate Yearly Payments (principal and interest), PI

   If L is the amount loaned, r is the interest rate, N is the loan duration, then the yearly payment for principal and interest, PI, is:

   $$\boxed{PI = L * \frac{r * (1 + r)^N}{(1 + r)^N - 1}}$$

9. Calculate Break Even Time for the Solar Energy System, T

   Depending on the tax rate, TR, of the user, the interest paid for any amount loaned for the purchase of the system may be tax deductible. If C_{EL} is the cost of electricity from the grid, and I_{EL} is the inflation rate of the electricity cost, then the break even time, T_{BE}, is when the cumulative cash flow of the amount invested to buy the system becomes zero.

   To estimate the annual cash flow and the cumulative cash flow up to the current time, n (in years), the following equations are used.

   At year n, the paid interest depends on the interest rate, r, and the remaining principal, RP:

   $$I_n = RP_{(n-1)} \cdot r$$

   The remaining principal is:

   $$RP_n = RP_{(n-1)} - (PI - I_n)$$

   where

   $$\mathrm{RP}_0 = L$$

   The tax savings, $S_{T,n}$, are:

   $$S_{T,n} = I_n \cdot TR$$

   The utility savings for the first year, $S_{U,1}$, are:

   $$S_{U,1} = Q \cdot 365 \cdot C_{EL} \cdot (1 + I_{EL}) / (1 - TR)$$

   The utility savings for the any other year n, $S_{U,n}$, are:

   $$S_{U,n} = S_{U,(n-1)} \cdot (1 + I_{EL})$$

   The annual cash flow, $CF_{A,n}$, is:

   $$CF_{A,n} = S_{U,n} + S_{T,n} - PI$$

   where

   $$CF_{A,0} = -(C_A - L)$$

   The cumulative cash flow, $CF_{C,n}$, is:

   $$CF_{C,n} = \sum_{i=1}^{n} CF_{A,i}$$

   The break even time, T_{BE}, is when:

   $$CF_{C,T_{BE}} = 0$$

Economic Analysis Results

A Microsoft Excel based tool has been developed to implement the previous procedure. The user inputs a number of data such as daily electricity requirements, cost of electric energy (utility) from the grid, utility inflation rate, percent of energy requirements to be met by the PV system, expected system efficiency, information about the system location, wattage cost, government credit, user tax rate, percent of system cost which is loaned, interest rate, and duration of loan. Then, the Excel tool estimates the system cost, cumulative cash flow, and break even time of the investment.

Fig. 3 summarizes the results of the economic analysis for a photovoltaic system that is designed to meet 70% of the energy requirements for a house in the Houston area.
Other user specified information is shown as Input Data. The Dependent Variables are shown as Results. The break even time is when the cumulative cash flow becomes zero.

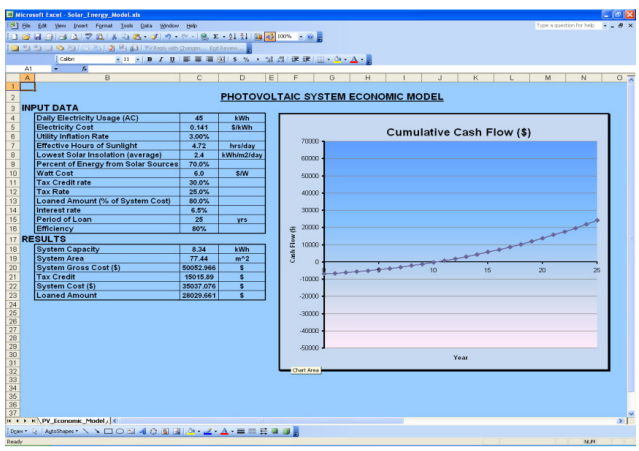


Fig. 3: Economic Analysis Results for a PV System in the Houston Area

The following observations can be made from the data in Fig. 3.

- The break even time is about 11.5 years for a photovoltaic system that meets 70% of the energy requirements of a house in the Houston area and when 80% of the total system cost is loaned.
- The system cost is $52,500.
- The cumulative cash flow is initially negative because of the 20% down-payment to purchase the system.

Effect of Location on Break Even Time

The solar insolation and effective sunlight hours vary from location to location. Thus, for the same energy requirements and cost parameters, the system location has an impact on the break even time.

The following figure refers to Phoenix, AZ with minimum monthly solar insolation of 2.72 kWh/m^2/day and 5.5 hours of effective sunlight.

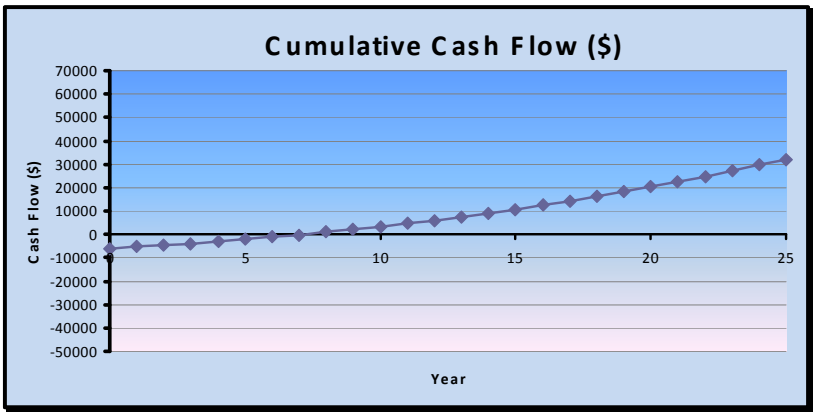


Fig. 4: Economic Analysis Results for a PV System in the Phoenix, AZ Area

When compared to Houston, the break even time is reduced from 11.5 years to 7.5 years due to the higher solar insolation and effective sunlight hours.

Effect of Loaned Amount on Break Even Time

Since the current cost of a PV system is considerable, the user may need to know what percent of the cost makes sense to borrow. For the same location, Houston, we will calculate the break even time when the loan amount is 80% or 100% of the system cost. The results for the case when 100% of the system cost is loaned are shown below.

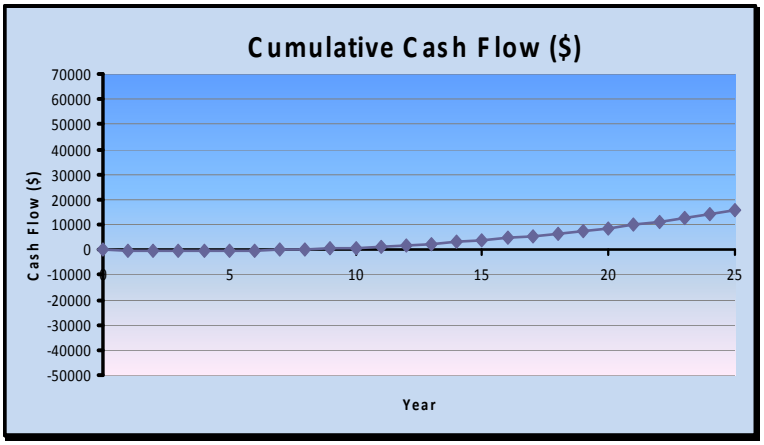
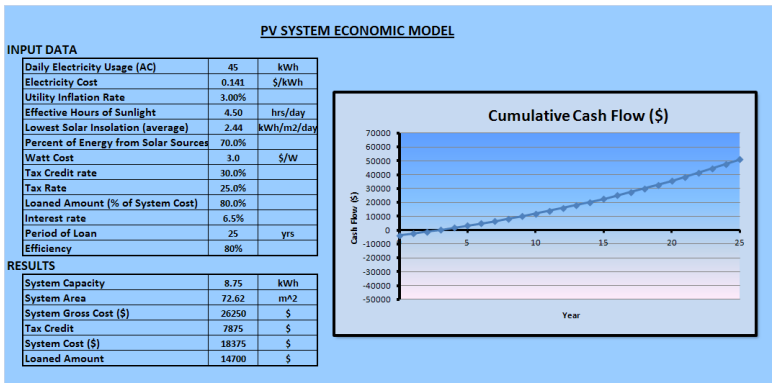


Fig. 5: Economic Analysis Results for the Houston Area, 100% of PV System cost is loaned

Compared to the case when the loan was 80% of the system cost, the break even time is reduced from 11.5 years to 7.0 years due to the tax credits received on the interest paid.

Effect of Wattage Cost

All previous analysis has been based on a wattage cost equal to $6.0/W. This is a typical value given the current state of PV systems technology. It would be interesting to know what the break even time would be if we were able to develop materials that lower the wattage cost to $3.0/W?



PV SYSTEM ECONOMIC MODEL

INPUT DATA

Daily Electricity Usage (AC)	45	kWh
Electricity Cost	0.141	$/kWh
Utility Inflation Rate	3.00%	
Effective Hours of Sunlight	4.50	hrs/day
Lowest Solar Insolation (average)	2.44	kWh/m2/day
Percent of Energy from Solar Sources	70.0%	
Watt Cost	3.0	$/W
Tax Credit rate	30.0%	
Tax Rate	25.0%	
Loaned Amount (% of System Cost)	80.0%	
Interest rate	6.5%	
Period of Loan	25	yrs
Efficiency	80%	

RESULTS

System Capacity	8.75	kWh
System Area	72.62	m^2
System Gross Cost ($)	26250	$
Tax Credit	7875	$
System Cost ($)	18375	$
Loaned Amount	14700	$

Fig. 6: Effect of Wattage Cost for the Houston Area, (other data same as Fig. 3)

Figure 6 shows that the break even time drops to 3.0 years from 11.5 years when the wattage cost drops from $6/W to $3/W. This is a significant result for it helps us focus our research efforts to develop materials which lead to lower wattage cost. This is in agreement with the statement by the National Academy for Engineering[1] that " *If the engineering challenges can be met for improving solar cells, reducing their costs, and providing efficient ways to use their electricity to create storable fuel, solar power will assert its superiority to fossil fuels as a sustainable motive force for civilization's continued prosperity.*"

III. PV System Operation: Modeling, Simulation and Optimization

The previous economic analysis shows that, in the long term, PV solar systems are a viable alternative to traditional energy sources. There has been interest in understanding and mathematically predicting the behavior of PV systems. A number of such models have been proposed over the years[3-5]. These models have been implemented using SPICE or Matlab[6]. Although such tools are used in power electronics and for engineering applications, it might be beneficial to use Microsoft Excel which is widely used for everyday calculations. Using Excel, a wider audience of students can be reached and may become more interested in better understanding and studying PV solar systems. Thus, this work implements a PV model in Microsoft Excel. Electric current to voltage (I-V) and power to voltage (P-V) curves are generated by the system. Results are compared with those reported for MSX-60 PV modules.

PV Cell Model

This section summarizes the PV model used in this work. The mathematical model describes the relationship between current and voltage of the cell based on solar insolation values and characteristics of the particular cell model.

The general equation is given below:

$$I = I_{SC,T_R} \cdot \frac{G}{G_R} + a \cdot (T - T_R) - I_0(T_R) \cdot \left(\frac{T}{T_R}\right)^{\frac{3}{n}} \cdot e^{\left[\frac{q \cdot E}{n \cdot K} \cdot \left(\frac{1}{T_R} - \frac{1}{T}\right)\right]} \cdot \left[e^{\frac{q \cdot (V + I \cdot R_s)}{n \cdot K \cdot T}} - 1 \right]$$

where:

$$I_0(T_R) = \frac{I_{SC,T_R}}{\left[e^{\frac{q \cdot V_{OC,T_R} / N_s}{n \cdot K \cdot T_R}} - 1 \right]}$$

The resulting PV cell power is:

$$P = V \cdot I$$

PV Cell Model User Interface

To calculate the power, P, we need to solve the previous two equations for the current, I, at a given voltage value, V. This is done numerically in Excel using the SOLVER method. The results are summarized in a user friendly interface as shown in Fig. 7.

PV MODULE CHARACTERISTICS				
	Parameter Name	Symbol	Value	Units
Electrical Characteristics	Number of Cells in Series	Ns	36	
	Number of Cells in Parallel	Np	1	
	Cell short circuit current	$I_{SC,TR}$	3.8	A
	Cell open circuit voltage	$V_{OC,TR}$	21.1	V
	Cell Ideal Factor	n	2.1	
	Reference Temperature	T_R	25	degC
	Reference Solar Insolation	G_R	1000	W/m^2
	Temperature Coefficient for Current	a	0.00065	
	Temperature Coefficient for Voltage	b	0.08	
Input Data	Cell Temperature	T	25	degC
	Solar Insolation	G	1000	W/m^2
	Module Voltage	V	15.00	V
Calculated Data	Module Current	I	3.80	A
	Power	P	57.00	W

CALCULATE

Fig. 7: User Interface of Generalized PV Cell Model

A number of parameters (Electrical Characteristics) are specific to the model of the photovoltaic cell and are provided by the manufacturer. In our study, they refer to the MSX-60 model manufactured by Solarex. In the Input Data, the user specifies the cell temperature and solar insolation. Then, for a desired voltage, V, the user can calculate the corresponding current, I, and the resulting power, P, by simply pressing the CALCULATE button. For instance, when the voltage is 15.0 V, the current is 3.8 A. This result is in close agreement with the numbers reported for MSX-60 [8].

Using this system, a number of studies can be conducted to better understand the system's operation and define conditions for optimum operation where the generated power is maximum.

Effect of Solar Insolation

Fig. 8 shows how the current and power as a function of voltage for different levels of insolation and constant cell temperature.

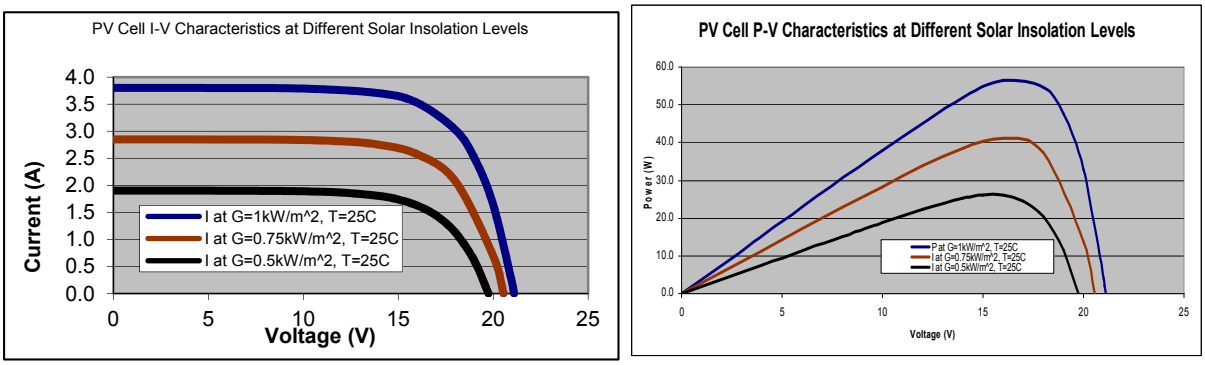


Fig. 8: PV Cell I-V and P-V Characteristics at Different Insolation Levels

Based on Fig. 8, at a given voltage value, the current increases as the solar insolation increases. There is a value of voltage which results in maximum power. This voltage value increases as the insolation level increases.

Effect of PV Cell Temperature

Fig. 9 shows how the current and power as a function of voltage for different levels of cell temperature and constant insolation.

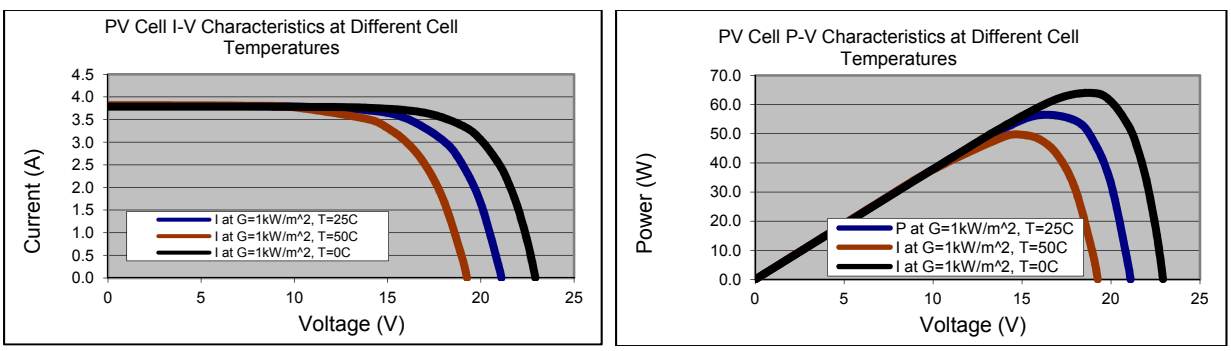


Fig. 9: PV Cell I-V and P-V Characteristics at Different Temperature Levels

At a given current value, the voltage increases as the solar cell temperature decreases. There is a value of voltage which results in maximum power. This voltage value increases as the cell temperature decreases.

IV. Educational Impact

This work combined economics and engineering to study a PV system. From an educational viewpoint, this work requires knowledge of and provides practice opportunities in economics concepts such as interest rate, break even time and cumulative cash flow. Furthermore, the implementation part of the economic analysis and simulation/optimization of the PV model

provided the students with many opportunities to learn about simple but effective computer platforms, such as Excel, and how to use them to address important engineering problems. The students learn quickly that technical decisions cannot be made without taking into consideration project economics. Last but not least, simulation tools can be used to analyze a problem and identify areas of improvement and where future research/development efforts should be directed to. In the case of the PV system, identifying methods to improve cell cost and efficiency is of high priority.

V. Conclusions

This paper has been concerned with the economic analysis as well as modeling, simulation and optimization of a Photo-voltaic (PV) system. The economic analysis, modeling, simulation, and optimization of the PV system have been implemented using Microsoft Excel. The economic analysis showed that solar energy is, in the long term, a viable alternative to other energy sources. To improve the economics of a PV system, research efforts must focus on developing lower cost and more efficient cell materials. The simulation and optimization results of the system are in agreement with experimental data reported for the MSX 60 PV system. Furthermore, to the best of the authors' knowledge, this is the first reported work where Microsoft Excel, an inexpensive and widely used tool, is used to model and optimize a PV system.

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